SharedPtr

SharedPtr

vector< smacc2::ISmaccState * >

SmaccStatus

map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >

recursive_mutex

atomic< StateMachineInternal Action >

map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >

map< std::string, std ::pair< std::function < std::string()>, boost ::any > >

vector< smacc2_msgs :msg::SmaccTransitionLogEntry >

SMRunMode

int64_t

SharedPtr

shared_ptr< smacc2 ::introspection::SmaccState Info >

shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >

string

vector< smacc2::introspection ::SmaccTransitionInfo >

vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >

type_info *

std::enable_shared _from_this< SmaccStateInfo >

map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >

map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >

map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >

int

#eventsLogPub_
#stateMachinePub_

#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_

#timer_

#currentState_

#status_msg_

#orthogonals_

-eventQueueMutex_
-m_mutex_

-stateMachineCurrentAction

-stateCallbackConnections_

-globalData_

-transitionLogHistory_

-runMode_

-stateSeqCounter_

#nh_

#currentStateInfo_

#stateMachineInfo_

+parentState_

+stateMachine_

+demangledStateName
+fullStateName

+transitions_

+children_

+tid_

+staticBehaviorInfo

+stateReactorsInfo

+eventGeneratorsInfo

+depth
+stateIndex_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

smacc2::introspection ::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

-statusPub_

SmaccFifoScheduler *

processor_handle

thread

ExecutionModel

vector< smacc2::ISmaccUpdatable * >

vector< std::vector < smacc2::ISmaccUpdatable * > >

atomic< int64_t >

double

atomic< bool >

mutex

vector< std::shared _ptr< smacc2::ISmaccClient > >

vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >

#node_

vector< std::shared _ptr< smacc2::StateReactor > >

vector< std::shared _ptr< smacc2::SmaccEventGenerator > >

shared_ptr< rclcpp ::Logger >

-scheduler_

-processorHandle_

-signalDetectorThread_

-executionModel_

-updatableClients_

-updatableStateElements_

-lastState_

-loop_rate_hz

-end_
-initialized_
-rosInitialized_

-smaccStateMachine_

-signalDetector_

-stateMachine_

-mutex_

#clients_

-clientBehaviors_

#stateReactors_

#eventGenerators_

#stateInfo_

#logger_

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

smacc2::ISmaccOrthogonal

+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()

smacc2::ISmaccState

+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

boost::signals2::signal
< Signature, optional
_last_value< typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature< n
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >

map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >

-stateMachine_

-stateMachine_

#parentState_

#components_

#stateMachine_

-nodeName_

-orthogonal_

-currentOrthogonal

-currentState

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

-client_change_state

-client_get_state_

-isShutdownRequested_

-eventMutex_

smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >

boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >

< void()>

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

SharedPtr

function< void()>

smacc2::SmaccSignal
< void()>

optional< std::future
< int > >

-postEventActivate
-postEventActiveShutdown
-postEventCleanup
-postEventConfigure
-postEventDeactivate
-postEventDestroy
-postEventInactive Shutdown_
-postEventOnActivate Error_
-postEventOnActivate Failure_
-postEventOnActivateSuccess_

-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_

+onTransitionActivate
+onTransitionActiveShutdown
+onTransitionCleanup
+onTransitionConfigure
+onTransitionCreate
+onTransitionDeactivate
+onTransitionDestroy
+onTransitionInactive Shutdown_
+onTransitionOnActivate Error_
+onTransitionOnActivate Failure_
...

-onFailure_
-onFinished_
-onSuccess_

-onEntryThread_
-onExitThread_

-lastTransitionEvent
-subscription_

cl_lifecycle_node::
CpLifecycleEventMonitor

+ onTransitionUnconfigured Shutdown_
+ onTransitionOnConfigureSuccess_
+ onTransitionOnConfigureFailure_
+ onTransitionOnConfigureError_
+ onTransitionOnActivateSuccess_
+ onTransitionOnDeactivate Success_
+ onTransitionOnDeactivate Failure_
+ onTransitionOnDeactivate Error_
+ onTransitionOnCleanupSuccess_
+ onTransitionOnCleanupFailure_
+ onTransitionOnCleanupError_
+ onTransitionOnShutdownSuccess_
+ onTransitionOnShutdownFailure_
+ onTransitionOnShutdownError_
+ onTransitionOnActiveShutdown Success_
+ onTransitionOnActiveShutdown Failure_
+ onTransitionOnActiveShutdown Error_
+ onTransitionOnErrorSuccess_
+ onTransitionOnErrorFailure_
+ onTransitionOnErrorError_
- postEventUnconfiguredShutdown_
- postEventOnConfigureSuccess_
- postEventOnConfigureFailure_
- postEventOnConfigureError_
- postEventOnDeactivateSuccess_
- postEventOnDeactivateFailure_
- postEventOnDeactivateError_
- postEventOnCleanupSuccess_
- postEventOnCleanupFailure_
- postEventOnCleanupError_
- postEventOnShutdownSuccess_
- postEventOnShutdownFailure_
- postEventOnShutdownError_
- postEventOnErrorSuccess_
- postEventOnErrorFailure_
+ CpLifecycleEventMonitor()
+ ~CpLifecycleEventMonitor()
+ setOwner()
+ getLastTransitionEvent()
+ onInitialize()
+ onStateOrthogonalAllocation()
- onTransitionEvent()

smacc2::SmaccAsyncClient Behavior

+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

string

-lastTransitionEvent_

-node_change_state _topic
-node_get_state _topic

-eventMonitor_

-eventMonitor_

cl_lifecycle_node::
ClLifecycleNode

+ ClLifecycleNode()
+ ~ClLifecycleNode()
+ onInitialize()
+ changeState()
+ configure()
+ activate()
+ deactivate()
+ cleanup()
+ shutdown()
+ destroy()
+ onStateOrthogonalAllocation()

-lifecycleNodeClient_

cl_lifecycle_node::
CbConfigure

+ CbConfigure()
+ ~CbConfigure()
+ onStateOrthogonalAllocation()
+ onEntry()